Wallabot Documentation

Wallabot is an AI-powered C2C second-hand marketplace built with a FastAPI microservices backend, a Vue 3 frontend, and an LLM-driven agentic subsystem for intelligent product classification and price recommendations.

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What is Wallabot?

Wallabot is a full-stack **C2C (consumer-to-consumer) second-hand marketplace** inspired by Wallapop. It allows registered users to list products, browse the catalogue, reserve items, and complete peer-to-peer purchases through an integrated digital wallet. The platform is augmented by an AI subsystem — the *Wallabot Agentic Service* — which provides intelligent product category suggestion and live market-based price estimation powered by LangChain LCEL, OpenAI GPT-4o-mini, and Tavily web search.

The project was developed across three sprints by team **BSPQ26-E9** as part of the Software Engineering course at the University of Deusto.

Sprint Deliverables

Sprint	Main deliverables
Sprint 1	User registration & login, product CRUD, basic catalogue browsing
Sprint 2	Digital wallet, product reservation with configurable timeout, atomic purchase flow, transaction history
Sprint 3	AI category suggestion agent, AI price recommendation agent, LangSmith observability tracing, user ratings system, public user profiles, full-text search on catalogue

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Key Features

User Management & Authentication

- Email and password registration with **bcrypt** password hashing
- JWT-based stateless authentication (HS256) shared across all microservices via a single `SECRET_KEY` environment variable — no inter-service token validation call per request
- Social login via **Google OAuth 2.0** and **Facebook OAuth** using the `authlib` library
- Account linking: a social account is bound to one provider and rejects conflicting link attempts (you cannot link the same email to two different providers)
- Token payload: `{"sub": "<user_email>", "exp": <unix_timestamp>}`

System Architecture

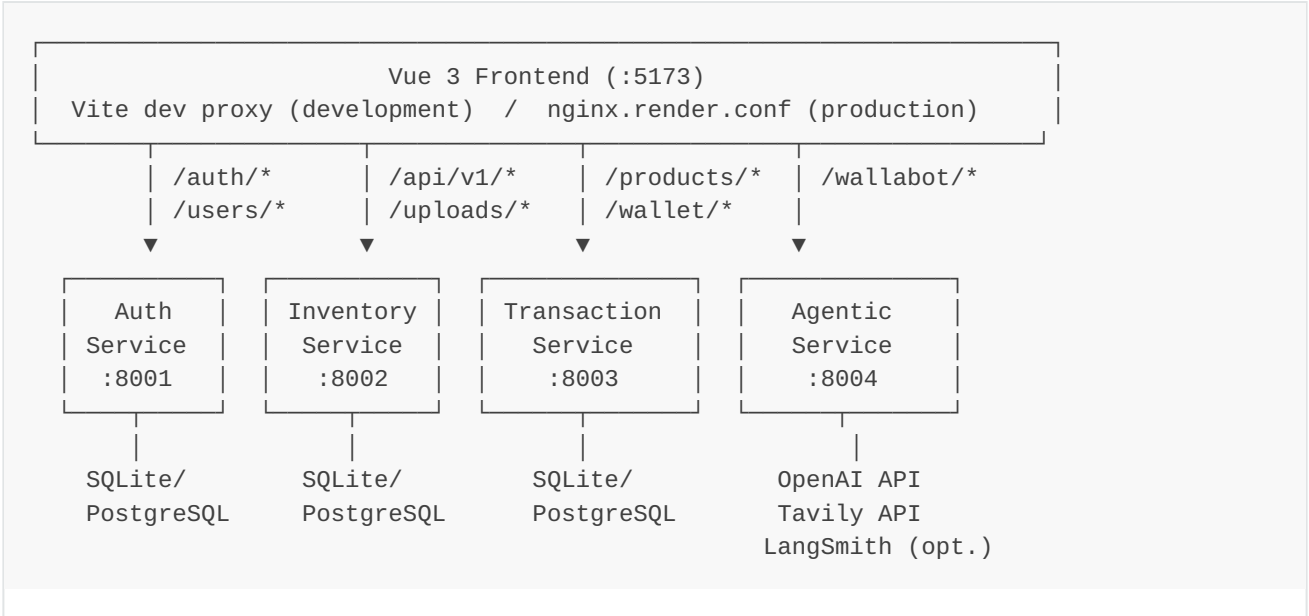
Design Philosophy

Wallabot follows a **microservices architecture** where each business domain is encapsulated in an independently deployable FastAPI service. Services share no databases and communicate only via HTTP REST. The Vue 3 frontend is the sole public entry point and routes every API call through a Vite proxy in development (or Nginx in production) to the appropriate service.

Key design decisions:

- **No shared database** — each service owns its schema; cross-service data access goes over HTTP with service-scoped JWTs
- **Stateless JWT validation** — every service validates tokens locally, eliminating a central auth round-trip per request
- **Append-only ledger** — wallet balance and product state history are immutable, audit-safe records
- **Graceful AI degradation** — agents always return a usable response even when the LLM provider is unreachable

High-Level Diagram



The Agentic Service has **no database** — every request is stateless and fully self-contained.

REST API Reference

Wallabot exposes its functionality through four independent HTTP REST services. This section documents every public endpoint, request/response schema, and error contract — the equivalent of an RMI interface specification.

All services accept and return **JSON** (`Content-Type: application/json`). Endpoints that require authentication expect a `Bearer` token in the `Authorization` header issued by the Auth Service.

Auth Service API

Base URL (local): `http://localhost:8001` **Base URL (production):** configured via `AUTH_SERVICE_URL` environment variable

The Auth Service owns user identity: registration, login, social OAuth, JWT issuance, public user profiles, and the ratings system.

Authentication

Most endpoints in other services require a JWT access token issued by this service. Pass it as a Bearer token in every request:

```
Authorization: Bearer <access_token>
```

Tokens are signed with `HS256` using the shared `SECRET_KEY` environment variable and contain `{"sub": "<user_email>"}` as the payload subject.

Health Check

`GET /health`

Returns the service liveness status. Used by Docker healthcheck probes.

Response 200

```
{ "status": "ok" }
```

Inventory Service API

Base URL (local): `http://localhost:8002`

The Inventory Service owns the product catalogue: creating, reading, updating, and deleting listings, plus image uploads. Products stored here represent the *listing* view; the Transaction Service maintains a separate copy for commerce state.

Health Check

`GET /health`

```
{ "status": "ok" }
```

Products

`GET /api/v1/products`

Returns a list of all products, optionally filtered. No authentication required.

Query parameters

Parameter	Type	Description
`state`	`Available` | `Reserved` | `Sold`	Filter by availability state
`category`	string (min 1 char)	Filter by category name (case-insensitive)
`min_price`	float (≥ 0)	Minimum product price in EUR
`max_price`	float (≥ 0)	Maximum product price in EUR
`condition`	`New` | `Like New` | `Good` | `Fair` | `Poor`	Filter by item condition
`q`	string (min 1 char)	Full-text keyword search across title and description

Response 200 — array of `ProductOut`

Transaction Service API

Base URL (local): `http://localhost:8003`

The Transaction Service owns the commerce domain. It manages a digital wallet per user, coordinates product reservations (with automatic timeout expiry), and processes purchases via atomic wallet debit operations. It also exposes transaction history.

All endpoints require authentication (`Authorization: Bearer <token>`).

Health Check

`GET /health`

```
{ "status": "ok", "service": "transaction-service" }
```

Wallet

`GET /wallet/balance`

Returns the authenticated user's current wallet balance in EUR.

Response 200

```
{ "balance": 250.00 }
```

`POST /wallet/deposit`

Adds funds to the authenticated user's wallet.

Request body

Agentic Service API (Wallabot)

Base URL (local): `http://localhost:8004`

The Agentic Service is the AI subsystem of Wallabot. It provides two intelligent endpoints that assist sellers when creating a product listing: automatic category classification and market-based price estimation.

All endpoints are **stateless** — each request is self-contained with no memory of prior calls. Authentication is **not required** to call these endpoints.

Health Check

`GET /health`

```
{ "status": "ok" }
```

Category Suggestion

`POST /wallabot/category`

Classifies a product into one of the caller-provided categories using a LangChain LCEL chain backed by **GPT-4o-mini**.

The agent selects the most appropriate existing category. It only proposes a new category name (setting `is_new_category: true`) when none of the provided options is a good fit.

On output validation failure the agent retries up to **3 times** with progressively detailed correction feedback. On LLM provider failure it returns the closest available category with `confidence: 0.0` rather than raising a 5xx error.

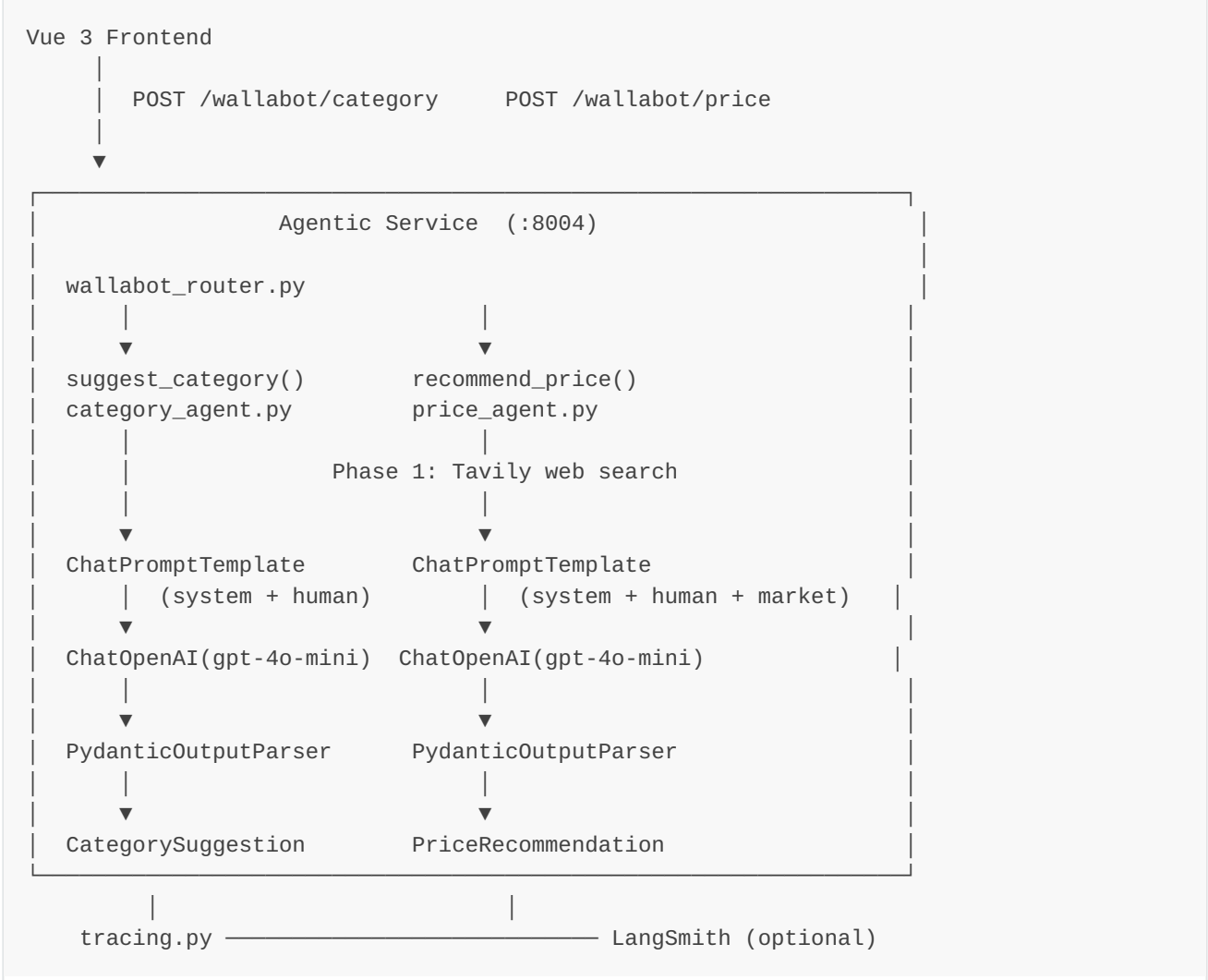
Wallabot AI Agent

The Wallabot Agentic Service (`backend/agentic-service/`) is the AI subsystem of the marketplace. It is an independent FastAPI microservice that exposes two intelligent endpoints to assist sellers when creating a product listing:

- `POST /wallabot/category` — automatic product category classification
- `POST /wallabot/price` — market-based price estimation in EUR

Both are powered by LangChain LCEL, OpenAI **GPT-4o-mini**, and the **Tavily** web search API. Every call is stateless — no session or memory is kept between requests.

Architecture Overview



Testing & Quality

Overview

Wallabot enforces a **50% minimum code coverage** threshold across all backend services. The CI pipeline fails if any service falls below this threshold. All four services exceed this threshold significantly — the current measured coverage, obtained by running the full test suite locally, is summarised below.

Coverage Summary

Service	Tests	Lines covered	Coverage
auth-service	34	331 / 415	**80%**
inventory-service	44	267 / 301	**89%**
transaction-service	57	511 / 619	**83%**
agentic-service	75	277 / 318	**87%**
Total	**210**	**1 386 / 1 653**	**84%**

> Coverage is measured with `pytest-cov` against each service's `app/` package. The 50% minimum threshold is enforced by `--cov-fail-under=50` in every test run.

Per-Service Breakdown

Auth Service — 80%

Run:

```
cd backend/auth-service
pytest --cov=app --cov-report=term-missing -q
```

Module	Stmts	Miss	Cover
`api/deps.py`	10	0	**100%**

DevOps & CI/CD

Docker Compose (Local Development)

The `docker-compose.yml` at the project root starts the full stack locally.

```
cp .env.example .env   # fill in secrets
docker compose up --build
```

Services and ports:

Service	Internal port	Exposed port
auth-service	8000	8001
inventory-service	8000	8002
transaction-service	8000	8003
agentic-service	8000	8004
frontend	5173	5173

All backend containers share the `wallabot-net` Docker network and the `wallabot-data` named volume that persists SQLite database files.

Environment Variables

Copy `.env.example` to `.env` and fill in:

Variable	Description
`SECRET_KEY`	JWT signing key (shared by all services)
`OPENAI_API_KEY`	Required by agentic-service for GPT-4o-mini
`TAVILY_API_KEY`	Required by agentic-service for web search
`SUPABASE_URL`	(Optional) Supabase project URL for PostgreSQL + Storage
`SUPABASE_SERVICE_ROLE_KEY`	(Optional) Supabase service-role key
`GOOGLE_CLIENT_ID` / `GOOGLE_CLIENT_SECRET`	(Optional) Google OAuth